

On Experiments in Design Science Research and Theory Development: A Literature Review

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Abstract

Experimental research is an established method in Information Systems Research and well suited for Design Science Research (DSR). Our review of 71 published research articles aims at contributing to the DSR body of knowledge by systematically collecting the dependent and independent variables experimentally tested. Here, we identify a lack of cumulative research in DSR as only very few theory constructs are (re-)used. The paper provides several implications for theory and an outlook on potentially fruitful avenues for future research.

1. Introduction

Experimental research is an established method and well suited for *Design Science Research* (DSR). According to Adelman [1] “experiments are, by far, the most common and commonly thought of empirical evaluation method” (p. 294). DSR which is concerned with the artificial, i.e. IT artifacts, can use experiments to systematically evaluate design alternatives and identify superior manifestations to evoke improvements [42]. A prominent illustration of the pervasion of experiments can be found in [28] who quotes experiments as one out of five evaluation methods.

Hypotheses are an important element of a scientific process and are subject to “experimental or observational tests” ([10], p. 409). A hypothesis specifies the relationship between independent and dependent variables [14]. A set of interrelated hypotheses is defined as a theory [14]. Therefore, beside the use of *principles* [28] and *methodologies* [43], the use of a *Design Theory* is needed to be able to support or reject hypotheses. In general, within an experimental setting a *Design Theory* and its *integrated hypotheses* include at least one dependent variable, and one or more independent variables which can be systematically manipulated within the experiment procedure. With focus on IS design

research, at least one independent variable can be manipulated through the design of an IT artifact. Conceptualizations have been made which are particularly suitable to support or reject hypotheses in DSR. A prevailing notation framed by Gregor [21] is the concept of an “exterior mode” which is concerned with “*why a certain feature (X) is superior in terms of a measurement (M) than a system without feature X*” (p. 7). Similar conceptualizations have been made by other researchers [7, 36].

A review of existing literature is essential to get a “*firm foundation for advancing knowledge [and] facilitates theory development*” ([58], p. xiii). However, systematic reviews of theoretical constructs in general or with regard to specific research methods such as experimental research are rare in the domain of Information Systems. Fjermestad and Hiltz [18] analyzed 200 experiments and offer a comprehensive overview on what has been studied including a detailed insight in the used variables and hypotheses. However, the study focuses only on Group Support Systems. More recently Larsen et al. [38] built upon the Inter-Nomological Network (INN) and offer an integrated tool which allows to build upon prior research by locating existing constructs and their measurements in the realm of behavioral science. Especially in terms of theory development and the use of existing knowledge, this lack is significant.

Against this background, the objective of this paper is to systematically reveal theoretical constructs in DSR endeavors empirically evaluated through experiments. Since current studies are focusing on conference proceedings [42], this study expands the current state of the art including research articles from IS journals. To address our research objective, this paper is guided by the following research question: *Which constructs have been a part of Design Theories and are studied in experimental settings?*

The remainder of this paper is structured as follows: In the following section, we present the research background of experiments in IS and Design Science. In Section 3, we present our research

framework. The research method is described afterwards (section 4) including the data collection and the coding procedure. Section 5 shows our findings and section 6 presents the discussion with some concluding remarks.

2. Research Background

2.1. Experiments in IS Research

Experiments are valuable instruments used for empirical research in Information Systems [12, 18]. Fjermestad and Hiltz [18], for example, present the intense use of experimental research in IS, particularly in the domain of Group Support Systems. Recently published contributions such as the development of a framework for the evaluation of experiments [42] further strengthen the use of experiments in IS research. One-fifth of all research in IS is conducted in a laboratory or field experiment [12, 18]. Therefore, we are focusing on these types exclusively. We exclude experiments which are conducted in computer and simulation environments, since they are implemented within processing units and they do not need participant involvement [11].

For the administration of experiments, researchers have to take into account several specific conditions and assumptions which enable them to test particular properties and characteristics. Firstly, control mechanisms are needed to test these properties and characteristics. Following Fjermestad and Hiltz [18], we refer to a controlled experiment if *“two or more conditions [are] deliberately created and contrasted, other variables were controlled in some manner, and there was at least one independent and one dependent variable, which was measured and statistically analyzed”* (p. 9). Secondly, an independent variable(s) is manipulated in a systematic manner. Yaremenko et al. [60] summarize *“one or more independent variables are manipulated to observe their effects on one or more dependent variables”* (p. 72). The similarity of both descriptions [18, 59] is to test descriptive, causal hypotheses about manipulable causes. To summarize, specific conditions and assumptions of experiments are two or more different groups within a designed and controlled setting [12]. Necessarily, one group is a treatment group, affected by stimuli, whereby the other group is controlled, without a stimulation. In this way, IS researchers are able to test and examine effects of the independent variable(s) on the dependent variable(s) in experimental settings [33, 12].

The distinction between a laboratory and a field experiment is the setting in which the experiment is conducted. Laboratory experiments are implemented

in a controlled environment where researchers are able to control the treatments and the setting itself. Field experiments are performed in the real world or a real world setting. Hence, an advantage of field experiments is generalizing real world situations in the experimental environment. Problems of field experiments are: (a) they are less controllable than laboratorial settings, and (b) the variation or manipulation of independent variables and factors or observation of these influences on the dependent variable is more difficult [12, 33].

The use of experiments in IS research has been successfully shown in many cases. Central to all these cases is studying an IT artifact which can be an E-commerce website, a computer interface, a website design, a virtual world or tools and software applications. Illustrative examples of studies within a laboratory setting can be found in [15] analyzing the effects of human images in website designs, in [26] studying the effects of social presence for the design of recommendation agents and in [56] analyzing the differences between an user-guided and a system-controlled decision aid.

Studies conducting field experiments can be found in [5] analyzing the competition between firms in online sponsored markets, [49] presenting the personalization-privacy paradox, [50] examining the effectiveness of website personalization or [51] studying the use of personal communication technologies (i.e. email and short message service).

Experiments are used to study a broad range of phenomena. Examples include the impact on usefulness [37, 51, 55], social presence [37, 61], trust [15, 26], performance [61, 9] or user behavior [50].

Conducting experiments allows conclusions to be drawn about superior design choices in terms of the dependent variable. For example, in [15] *“the high-human condition is perceived as most favorable and resulted in image appeal and social presence for users”* (p. 554), and thus, it is superior to a no-human image in terms of image appeal, social presence and trust.

With the manipulation of independent variables, a set of design alternatives is addressed at the same time, since an explanation is given about why a specific design alternative is more favorable than another. Explaining “why” design alternatives are favorable or superior to others, the question is justified whether this kind of research follows the Design Science Research or not.

2.2 Design Theory

Rooted in the distinction between the science of the natural and the science of the artificial, there are

primarily two research paradigms in IS research: *Behavioral Science* and *Design Science* [39]. Both paradigms are broadly represented in IS research whereby a growing number of researchers are concerned with *Design Science* (see for instance [27]). Influential contributions, such as Design Science methodologies [43] or the seven guidelines by Hevner et al. [28] are further boosting *Design Science Research*.

The use of experiments is a central method for the evaluation of artifacts in Design Science Research [28, 44]. However, little research has been done so far on the relationship between DSR and the use of experiments (e.g. [42]). In fact, there are only few studies addressing both aspects, including the work of Fjermestad and Hiltz [18] analyzing theoretical constructs used in experimental settings and Mettler et al. [42] developing an evaluation framework for DSR and experiments. Based on the growing interest in DSR, we consider this as a gap. Especially, studies with a strong integration of theory and the evaluation method are needed to further advance DSR. This is of special significance since several researchers emphasize the importance of theory in DSR (e.g. 54, 20, 22). Especially within the discussion about *Design Science*, arguments are made that the pure focus on IT artifacts is too narrow for a social-technical discipline [40, 11, 35]. Along with concerns about the theory in DSR it is necessary to clarify what an “IS Design Theory (ISDT)” is. For this paper and for experimental research in general, an ISDT is needed which is closer to the nature of experiments in a way that ISDT includes (controlled) variables which can be manipulated within the administration of the experiment and tested empirically. Thus, we need an understanding of Design Science which is not only focused on the IT artifact itself, but also on a Design Theory.

There are various approaches to explain the role of theory in DSR. Based on Simon’s work about *Science of the artificial* [47], Gregor [21] is arguing for a dualistic point of view in theorizing, proposing the interior and the exterior mode. Defined by Gregor “*the interior mode is where theorizing is done to produce theory for design and action, with prescriptive statements about how artifacts can be designed, developed and brought into being*” (p. 7), whereby the “*exterior mode includes other types of theory, which aim primarily at analyzing, describing and predicting what happens as artifacts exist and are used in their external environment*” (p. 7). Similar arguments are made by Baskerville and Pries-Heje [7] who split *Design Theory* in a *Design Practice Theory* and an *Explanatory Design Theory*. The Design Practice Theory explains how to build an IT artifact; the

Explanatory Design Theory explains why specific characteristics are used within the process of building an artifact and, thus, are able “*to explain range of phenomena rather than a specific instance of a problem*” (p. 273). A similar point of view is stated by Kuechler and Vaishnavi [35] who propose the notion of a “*Design Relevant Explanatory / Predictive Theory*” (DREPT) which explains “*how and why the artifact functions as it does; specifically, it explains how novel artifact design features have the effects they do*” (p. 397). From all these various notions, we can derive crucial aspects in terms of a Design Theory. Firstly, to explain a “*range of phenomena*” [44], it is necessary to generalize which can be achieved by using testable hypotheses. Secondly, following Gregor [21], hypotheses should be framed as “*a system with feature X will perform better on measure M than a system without feature X*” (p. 7). Thirdly, referring to the science of the artificial [47], at least one feature is included in an IT artifact. To put it simply, a Design Theory is supposed to inform the designer of an IT artifact about which features to include and for what purpose. Bringing these aspects together, we propose the following definition of a Design Theory:

A Design Theory is a collection of two or more connected hypotheses explaining desired effects (dependent variable), whereby at least one independent variable (feature) can be designed through manipulation of an IT artifact.

The use of this definition offers two essential benefits: (1) the emphasis on the use of hypotheses by the given definition is close to natural science. Therefore, it supports the integration of theories from behavioral science (e.g. theories from sociology and social science). (2) without the constraint that all independent variables need to be within the IT artifact (i.e. features), this definition allows a hybrid form of research. (3) with the emphasis on the use of hypotheses, empirical evaluation (falsification) is possible [14].

3. Research Framework

In order to analyze constructs which have been studied in experimental settings, we briefly introduce our research framework (see Figure 1) which will be the foundation of our structured analysis (see 4.2 Data Coding). The research framework is composed of the IS Design Theory elements and elements relating to the experimental setting. For the IS Design Theory part, we include theory constructs (i.e. independent and dependent variables). Furthermore, we differentiate between independent variables which are manipulated within an IT artifact (e.g. human image on a website [15]), and variables which are outside the

IT artifact (e.g. security behavior, self-efficacy [4]). Next, we integrate elements concerning the experimental setting. These include the description of the experimental setting (e.g. randomized setting), the form of an experiment (e.g. laboratory experiment) and the subject group (e.g. students). Finally, the measurement is included. We distinguish the following forms of measurement: the measurement method for the experiment itself (e.g. surveys), and the measurement for the data analysis (e.g. statistical methods). Besides the use of single measurements, there is considerable research using multi-measurement approaches, combining questionnaires with interviews and eye-tracking methods [15] or electroencephalography with questionnaires [23].

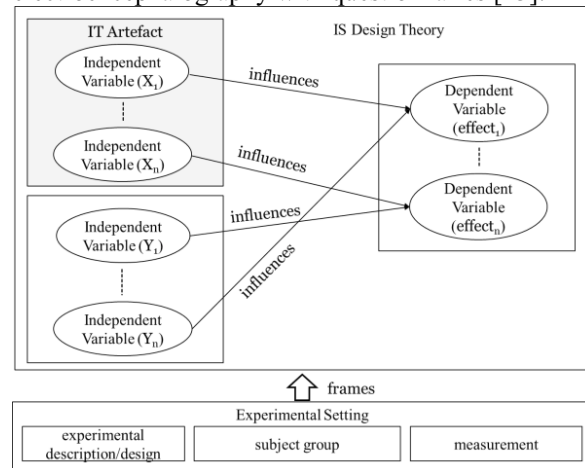


Figure 1: Research framework

4. Research Method

4.1 Data Collection

To identify relevant articles, we conducted a literature search [53]. To gain a suitable amount of data for the exploration of theoretical constructs in Information Systems Research we selected two IS journals, i.e. *Management Information Systems Quarterly (MISQ)* and *Information Systems Research (ISR)*, because they include IS leading scholars' works and are widely distributed in IS research.

Using the keywords “*experiment(s)*” and “*experimental*” between 2005 and 2015 within titles, abstracts and in author keywords we were able to collect a significant number of papers on experiments for an in-depth analysis. Our search yielded 104 papers in total (50 ISR and 54 MISQ). We excluded from our search “issues and opinions” as well as “editorial notes” papers.

Next, we studied all the articles and excluded those that are not using experiments to test hypotheses or match our understanding of an experiment in an

experimental setting. The remaining 71 papers were analyzed in detail.

4.2 Data Coding

The data coding procedure was conducted as follows: Firstly, we read five random papers of each journal, to identify aspects which are relevant for our research question. We categorized all aspects and together with elements of a Design Theory (e.g. independent variable) we integrated them in an initial coding scheme.

Secondly, using the initial coding scheme, we read the residual papers individually and adjusted the scheme consequently. After each paper has been read by the first and second author, discrepancies were identified and discussed with a third researcher to adapt the coding scheme and come to a final agreement. The final scheme is illustrated in figure 1. With the creation of an initial coding scheme and an iterative procedure, we followed the Template Based Approach [34]. We chose this approach because it allows us to start with an initial coding scheme based on our previous work and adapt it within the coding procedure.

Finally, we analyzed the 62 dependent and 56 independent variables (see Appendix) for relationships. The process of relating similar codes to each other is in line with axial coding suggested by Grounded Theory [13]. This resulted in four subcategories (c.f. table 1) which can be used as a starting point for researchers searching for system-related or user-related knowledge.

- *System-design variables* contain all variables that influence the design of an IT artifact (e.g. websites, displays, images or other visual design features). Examples are E-Images [19] or navigation systems [57].
- *User-design variables* include all variables that are influencing users' behavior and attitude directly or indirectly. Examples are content quality [41] or the amount of provided information [31].
- *System-related effects* include all aspects looping back to the system to enhance its functions or usability. Examples are performance [9] or form response [2].
- *User-related effects* comprise all outcomes of the experimental manipulation on the side of the system user and contain, for example, changes in attitude (e.g. trust [15]), intention (e.g. intention to return [57]), behavior (e.g. bidding behavior [30]), or capability (e.g. learning outcomes [25]).

Independent Variables	
<i>System-design variables</i>	<i>User-design variables</i>

Dependent Variables	
System-related effects	User-related effects

Table 1: Categorization of the variables

In terms of the experimental setting, we identified the aspects of the experimental description and design including general aspects of experiments (e.g. field vs. laboratory experiments), the subject group and the used measurement approaches. Furthermore, we identified all independent and dependent variables systematically. Following our definition of Design Science (defining criterion: Design Theory), we classified research articles having an independent variable within an IT artifact as Design Science. If an article deals with theory in which no variable is subject to manipulation through an IT artifact, we classify it as Behavioral Science.

5. Findings

Our study reveals that much of the experimental research in IS features elements of Design Theory. Following our definition, an overall of 44 (62%) of the reviewed articles are using a Design Theory. With regard to the variables, 57% of the independent variables are system-design related and 44% are user-design variables. Furthermore, 24% of the dependent variables (effects) are system-related, whereby 76% are user-related. Table 2 illustrates an extract of our findings; table 4 and table 5 in the Appendix include a comprehensive overview of our findings.

Independent Variables	
System-design variables (57%)	User-design variables (43%)
Display order [17] Human image [15] E-Image [19]	Preferred distance [29] Feedback [6] Privacy [52]
Dependent Variables	
System-related effects (24%)	User-related effects (76%)
Query performance [3] Form response [2] Ad performance [5]	Consumer learning [48] Bidding behavior [30] Intention to return [32]

Table 2: Independent and dependent variables

We further derived major findings in terms of the organization and conduction of experiments (c.f. Table 3). Firstly, 80% of the reviewed articles are using a laboratory setting. Only 16% are conducted in a field experiment and 4% are applying a mixed approach. Almost all of the experiments are conducted in a randomized manner (98%), only 2% of the reviewed articles are applying quasi-experiments (i.e. participant assignment done by the researchers). In terms of the experimental design, the majority of

research relies on a between-subject design (73%), 11% are using a within-subject design and the remaining 16% are using a hybrid approach.

Secondly, in a majority of the reviewed experiments, research students (undergraduate as well as graduate students) are participating in the experimental procedure (73%). Only about every fourth study is incorporating specific subject groups such as university staff, workers, managers, software developers or IT users (23%) and only 2% is using both.

Thirdly, the most commonly used method for the evaluation are questionnaire-based surveys (65%). In addition, other methodological approaches are multi-method approaches in which the combination of two or more methods is applied (27%). Typical methods used in combination are: questionnaires, interviews, eye-tracking methods or neurophysiological (Functional Magnetic Resonance Imaging, Electroencephalography, Skin Conductance Response etc.). A few researchers utilize other forms such as online data (9%). For the analysis of the data 21 articles are using descriptive statistics, 24 are applying an ANOVA analysis, 15 articles are using MANOVA, 3 articles are including ANCOVA method for their analysis and 4 papers MANCOVA. To check distinct models in the research model the Partial Least Squares (PLS) method is used 12 times and the structural equation modeling 5 times. In 18 articles pre-test and statistical analysis is made by other methods (e. g. chi-square, hypothesis (t)-test etc.). All over, in 38 (86%) paper the researcher conducted a multi-analysis approach.

Experimental Setting	Occurrence
Laboratory Experiment	80%
Field Experiment	16%
Combination of both	4%
Experimental Description	
Randomized Experiment	98%
Quasi-Experiment	2%
Experimental Design	
Between-subject Design	73%
Within-subject Design	11%
Combination of both	16%
Subject Group	
Students	73%
Target Group (e.g. Managers)	23%
Mixed	4%
Measurement Method	
Questionnaires	64%
Multi Method Approaches	27%
Other (e.g. Online Data)	9%

Table 3: Overview experimental information

6. Discussion

Our systematic analysis of constructs studied in experimental settings reveals a broad spectrum of knowledge used in system-related and user-related research. Based on this insight, we derive four major results, which we expose in the following.

Research paradigm in experimental settings: More than half of the reviewed research articles match our definition of a Design Theory and, thus, can be classified as Design Science. This is remarkable, since some researchers would understand themselves as researchers in the field of behavioral science (see for instance [8]). The major reason for this result rests upon our understanding and definition of a Design Theory, since we strictly analyzed whether there is an independent variable which is manipulated within an IT artifact or not. Therefore, our understanding of Design Science, Design Theory and the question of why specific components are constructed into an IT artifact is in line with the *exterior mode* [21], with the *Explanatory Design Theory* [7], and with the *Design-Relevant Explanatory/ Predictive Theory* [35].

On the use of experiments: A majority of research is using laboratory experiments (80%), having the advantage of a controlled and artificial environment where particular treatments can be tested with diverse subject groups [12]. However, in contrast to field experiments, laboratory experiments are more distanced to real world scenarios than field experiments. Especially in terms of studying technology in everyday life, such as Consumer IT, an artificial environment might be a disadvantage. Our findings imply that questionnaire-based surveys are the most favored method to evaluate experiments. One reason might be a broad knowledge in terms of the method and its application. However, recent developments in the domain of neurophysiology offers new methods (e.g. [45] or [46]) which seems to be more reliable in terms of their evaluation [16]. Until now, only a few researchers have combined traditional methods (e.g. questionnaires) with neurophysiological approaches (e.g. electroencephalography) [24].

Area of use: The differentiation between system-related variables and user-related variables shows that both areas are relevant for IS research and are studied by using experimental settings. However, the majority of effects are concerned about user-related effects. The reason for this emphasis could be due to the background of the researcher, e.g. from Human-Computer-Interaction or with a focus on End-User Research. Thus, we have less experimental work with system-related effects. On the one hand this is strengthening user-related research, but on the other

hand it weakens Design Science endeavors which are concerned with system-related aspects.

Theoretical constructs in experimental settings; a matter of integration: There is a broad spectrum of variables empirically tested in experimental settings. However, only a small number of concrete variables are used or reused by more than one article. This heterogenic and poorly integrated map is critical since research pursues to build upon existing knowledge [58, 53]. One reason for this may lie in the enormous amount of constructs empirically studied in IS research. Addressing this issue, we provide an initial step to support future research to build their endeavors on “*the shoulder of giants*” [53] by systematically consolidate system-related and user-related knowledge.

Limitations and outlook: Our study has certain limitations. Firstly, due to the choice of two IS journals and the restricted time-period, we are not able to capture a comprehensive review of all constructs studied in experimental settings. We rather focused on an in-depth review of current articles. Further research could enrich our contribution by including more papers of IS journals and conferences to complement our initial work. Secondly, the differentiation between system-related and user-related constructs is only a first step towards a systematic representation of the constructs. Further research is needed to analyze the data in a more detailed way. With regard to the broad spectrum of theories used in IS research, methods such as theoretical coding as part of Grounded Theory [13] could help to go one step further and identify the connection between the revealed constructs and existing theories.

Besides addressing the limitations, potential future research could cover the following points: Apart from our approach to differentiate between system-related and user-related aspects, new conceptualizations can be developed to support specific areas of interest such as mobile technologies. Furthermore, we encourage researchers to continue the discussion about DSR and the role of theory, especially in terms of common features of both paradigms.

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Appendix

	#	Independent Variables	Author	Year	Outlet
System-design Variables	1	scripts (guided or non-guided)	Bera et al.	2014	ISR
	1	user-system interaction mode (USIM)	Wang & Benbasat	2013	ISR
	1	use of guided, unguided ontologies	Bera et al.	2011	MISQ
	3	familiarity, delay, breadth (website design variables)	Galletta et al.	2006	ISR
	1	display condition	Dabbish & Kraut	2008	ISR
	2	visual design features of webpage (order, visual complexity)	Deng & Poole	2010	MISQ
	1	product presentation format	Jiang & Benbasat	2007	MISQ
	1	trade-off transparency	Xu et al.	2014	MISQ
	1	decision aids (screening support)	Tan et al.	2010	ISR
	1	goal information	Kuechler & Vaishnavi	2006	MISQ
	1	strengths of VR interfaces (media richness, interactivity, telepresence)	Suh & Lee	2005	MISQ
	1	display order	Dou et al.	2010	MISQ
	1	initial freedom of choice (for interfaces)	Murray & Häubl	2011	MISQ
	2	navigation support, communication support	Zhu et al.	2010	ISR
	1	human image	Cyr et al.	2009	MISQ
	1	recommendation agents (personalization)	Komiak & Benbasat	2006	MISQ
	1	E-Image	Gregg & Walczak	2008	MISQ
	2	presence of image and text image, image motion	Lee et al.	2012	MISQ
	2	recommendation agents (weighted, cutoff)	Lee & Benbasat	2011	ISR
	1	ABDSS Use (attribute-based decision support system)	Kamis et al.	2008	MISQ
	2	recommendation agents, consumer reviews	Kumar & Benbasat	2006	ISR
	1	web stimulus (personalized offers)	Tam & Ho	2006	MISQ
	1	navigation systems	Webster & Ahuja	2006	MISQ
	1	set size	Tam & Ho	2005	ISR
	1	decision strategy support (decision)	Wang & Benbasat	2009	MISQ
User-design Variables	1	advertiser strategy	Animesh et al.	2011	ISR
	1	portal affiliation (display)	Sia et al.	2009	MISQ
	1	characteristics of shared information	Hinz & Spann	2008	ISR
	1	policy threshold (adaptive/fixed)	Hinz et al.	2011	MISQ
	1	feedback	Arnold et al.	2006	MISQ
	1	multipart pricing scheme	Kim & Bockstedt	2013	ISR
	1	personal communication technology (SMS/eMail)	Tan et al.	2014	ISR
	3	content quality, source expertise, validation	Meservy et al.	2014	ISR
	1	technology mediation for learning	Gupta & Bostrom	2013	ISR
	1	software design patterns	Mangalaraj et al.	2014	MISQ
	1	advertisements	Sutanto et al.	2013	MISQ
	3	claims, data and backing	Kim & Benabsat	2006	ISR
	2	amount of information; direction of attention	Hong et al.	2013	MISQ
	1	piracy prevention	Raghu et al.	2009	ISR
	1	anchor (recommender)	Adomavicius et al.	2013	ISR
	1	evaluation support	Smith et al.	2011	ISR
	1	preference distance	Hildebrand et al.	2013	ISR
	1	ontological foundation	Alleen & March	2006	MISQ
	1	privacy (privacy policy)	Tsai et al.	2011	ISR

Table 4: Independent variables

	Dependent Variable(s)	Author	Year	Outlet
System-related Effects	privacy safe-application	Sutanto et al.	2013	MISQ
	variety of self-designs	Hildebrand et al.	2013	ISR
	adherence	Arnold et al.	2006	MISQ
	explanation access	Arnold et al.	2006	MISQ
	query performance (accuracy, confidence, prediction)	Alleen & March	2006	MISQ
	ad performance (click through rate)	Animesh et al.	2011	ISR
	participants performance in problem-solving	Bera et al.	2014	ISR
	performance	Bera et al.	2011	MISQ
		Galletta et al.	2006	ISR
	joint performance (of awareness information)	Dabbish & Kraut	2008	ISR
	effort variable (no. of iterations)	Wang & Benbasat	2013	ISR
	coordination performance	Zhu et al.	2010	ISR
	form response	Adomavicius et al.	2013	ISR
	decision	Kim & Bockstedt	2013	ISR
	size of bundle	Kim & Bockstedt	2013	ISR
User-related Effects	behavioral intentions / use intentions	Galletta et al.	2006	ISR
		Lee et al.	2012	MISQ
	behavior (usefulness)	Tam & Ho	2005	ISR
	perceived usefulness	Smith et al.	2011	ISR
		Kumar & Benbasat	2006	ISR
	perceived understanding	Bera et al.	2011	MISQ
	perceived ease of use	Bera et al.	2011	MISQ
	usefulness (purchases)	Tsai et al.	2011	ISR
	online behavior (approach-avoidance behavior toward website)	Deng & Poole	2010	MISQ
	preference of the incumbent interface	Murray & Häubl	2011	MISQ
	quality variable (perceived system quality)	Wang & Benbasat	2013	ISR
	bidding behavior	Hinz & Spann	2008	ISR
	intentions to return	Jiang & Benbasat	2007	MISQ
		Webster & Ahuja	2006	MISQ
		Kamis et al.	2008	MISQ
	willingness to pay / transact	Raghu et al.	2009	ISR
		Gregg & Walczak	2008	MISQ
		Kamis et al.	2008	MISQ
	intention to use recommendation agents / decision aids	Lee & Benbasat	2011	ISR
		Wang & Benbasat	2009	MISQ
		Komiak & Benbasat	2006	MISQ
	advice acceptance	Wang & Benbasat	2013	ISR
	effort variable (perceived cognitive effort)	Wang & Benbasat	2013	ISR
	decision quality	Kuechler & Vaishnavi	2006	MISQ
	quality of solution	Mangalaraj et al.	2014	MISQ
	task completion time	Mangalaraj et al.	2014	MISQ
	task satisfaction	Mangalaraj et al.	2014	MISQ
	customer satisfaction	Hinz et al.	2011	MISQ
	negative affective toward the wait	Hong et al.	2013	MISQ
	profit	Hinz et al.	2011	MISQ
	actual buy	Sia et al.	2009	MISQ
	price premium	Gregg & Walczak	2008	MISQ
	consumer learning	Suh & Lee	2005	MISQ
	learning outcomes	Gupta & Bostrom	2013	ISR
	enhanced memory of unknown brand	Dou et al.	2010	MISQ
	enhanced evaluation of unknown brand along primed attribute	Dou et al.	2010	MISQ
	trust	Cyr et al.	2009	MISQ
	trusting beliefs	Kim & Benabsat	2006	ISR
	social presence	Zhu et al.	2010	ISR
		Kumar & Benbasat	2006	ISR
	retention in choice set	Meservy et al.	2014	ISR
	perceived coupon value	Tan et al.	2014	ISR
	redemption of coupons	Tan et al.	2014	ISR
	forwarding of coupons	Tan et al.	2014	ISR
	satisfaction with self-design	Hildebrand et al.	2013	ISR
	perceived quickness of the wait	Hong et al.	2013	MISQ
	evaluation	Tam & Ho	2006	MISQ

Table 5: Dependent variables